Aging in Autism
Brain, Motor, and Memory Markers of Neurodegenerative Risks

Presenter:
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Autistic Adults show signs of increased **aging vulnerabilities**

BACKGROUND

Aging autistic adults are a growing and understudied population. Daily functioning depends on integrated motor, language, and executive capacities; yet, lifelong challenges in these areas are common for autistic people. Further, cortical brain networks scaffold these domains and are sensitive to age-related decline.
Middle-aged and older autistic adults are **2.6 times more likely** to be diagnosed with early-onset **Alzheimer's Disease (AD)** and **6 times more likely** to be diagnosed with **Parkinson's Disease (PD)** than non-autistic adults. Cognitive and motore challenges are considered comorbid symptoms and are commonly experienced from childhood and persist and worsen into middle age. However, very little is known about the biological underpinnings of increased AD and PD risk in autistic adults.

METHODS

Participants: Autistic and non-autistic adults without Intellectual Disability spanning early adulthood through older age; autism diagnosis confirmed (e.g., ADOS-2). Participants ≥40 completed biennial follow-up.

Approaches:
Aim 1 MRI + behavior: Whole-brain cortical morphometry from MRI (surface area primary; thickness/volume) with longitudinal mixed-effects models, linked to executive function tasks (Trails, Stroop), verbal fluency (Animal Naming), and fine motor speed (Finger Tapping).

Aim 2 Parkinsonism risk: PD polygenic risk scoring and clinician-rated Movement Disorders Society-Unified Parkinson's Diseae Rating Scale (MDS-UPDRS) Part III.

Aim 3 AD vulnerability: Ultra-sensitive plasma proteomics (NULISA™) for AD-related markers and exploratory proteins; longitudinal verbal long-term memory (Auditory Verbal Learning Test; AVLT).

Aim 1

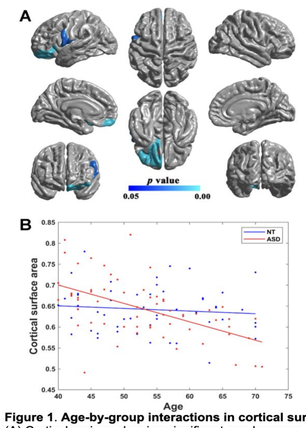

Figure 1. Age-by-group interactions in cortical surface area. (A) Cortical regions showing significant age-by-group interactions in cortical surface area on the white matter surface. Results are cluster-level corrected for multiple comparisons using random field theory (RFT). The color bar indicates RFT-corrected p values, with significance set at $p < 0.05$. (B) scatterplot illustrating the age-by-group interaction in the left orbitofrontal cortex and pars orbitalis/triangularis.

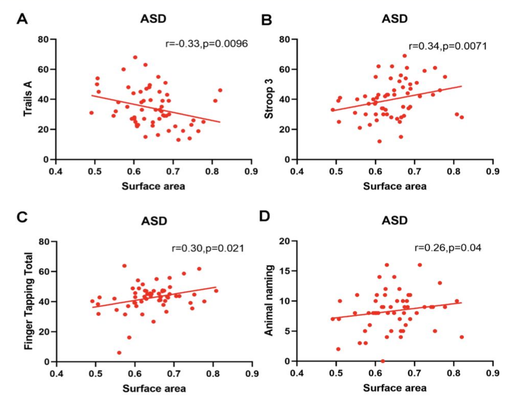

Figure 2. Correlations between orbitofrontal surface area and behavioral scores for (A) Trails A total time, (B) Stroop Color-Word Interference total words, (C) Finger tapping total score across three trials (average of dominant and non-dominant hand), and (D) Animal naming total words.

Aim 2

Figure 1. Group differences in Parkinson's Disease polygenic scores (PD-PGS)

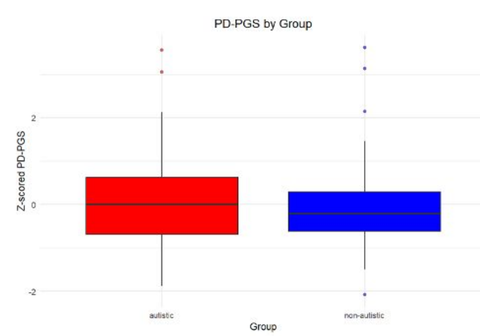

Figure 2. Group differences in parkinsonism symptom severity (MDS-UPDRS Part III)

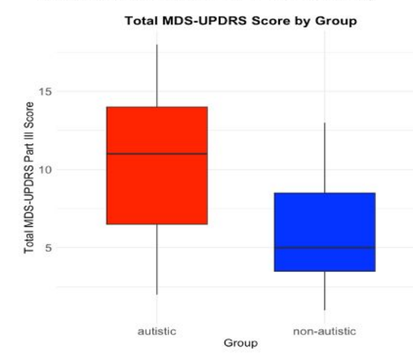

Figure 1. Autistic adults exhibited significantly larger PD-PGS ($t(134)=2.98$, $p=0.003$) than non-autistic adults.
Figure 2. Autistic adults showed significantly higher parkinsonism symptoms compared to non-autistic adults on the MDS-UPDRS Part III ($t(25)=-2.46$, $p=0.021$), with a large effect size (Cohen's $d=0.98$).

Aim 3

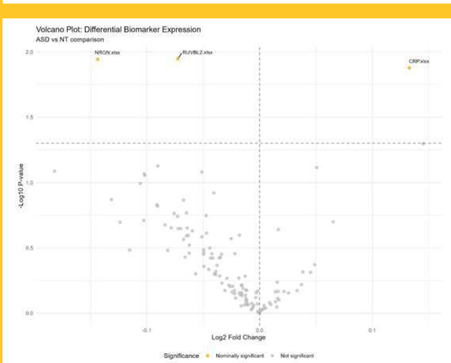

Figure 1. Differential biomarker expression reflected that C-reactive protein (CRP) was upregulated (log_2 fold change=0.133, FDR $p=0.013$), while neurogranin (NRGN; log_2 fold change=-0.144, FDR $p=0.011$) and RuvB-like 2 (RUVBL2; log_2 fold change=-0.073, FDR $p=0.011$) were downregulated.

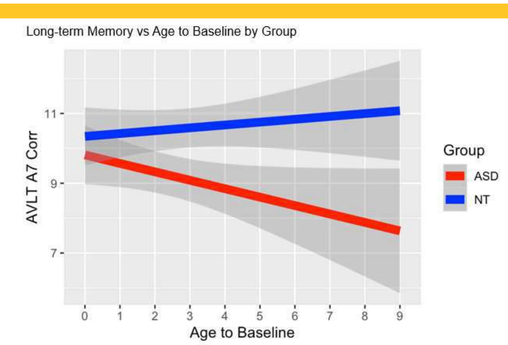

Figure 2. Multilevel modeling revealed a group by time interaction ($p=0.034$), with the ASD group demonstrating a steeper negative decline in scores over time.

RESULTS

Brain structure + behavior
•Autistic adults showed greater age-related surface area loss in left frontal (orbitofrontal / inferior frontal) and motor (precentral) regions.
•Smaller frontal/motor surface area in autistic adults related to poorer executive performance, slower finger tapping, and lower semantic fluency.

Parkinsonism
•Autistic adults showed higher PD polygenic risk and greater clinician-rated parkinsonism severity than non-autistic adults.

Memory + blood biomarkers
•Autistic adults showed steeper longitudinal decline in long-term verbal memory.
•Plasma proteomics show elevated Aβ42 and group differences in exploratory proteins (e.g., CRP↑, NRGN↓, RUVBL2↓), with modest individual discrimination.

CONCLUSION

Across MRI, genomic, and proteomic modalities, findings point to frontal–motor system vulnerability and elevated neurodegenerative risk in aging autism.

ACKNOWLEDGEMENTS

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